OFFICIAL TRAINING PROPOSAL | HANDS-ON STEM KIT | AGES 8–12

ILLUSTRATED MANUAL & BONUS PROJECTS

Hands-On Robotics & Hardware STEM Masterclass (Ages 8–12)

Educational Aim: To make students familiar with fundamental concepts of electronics and circuits, preparing them with confidence for future robotics, automation, and advanced engineering courses.

📘 **Comprehensive User Manual Included:** Each kit includes a detailed, child-friendly illustrated guidebook featuring step-by-step wiring guides, safety rules, and instructions for **many additional DIY projects** that can be built at home using these exact components!

Offered By: CS Technologies

Instructional & Management Team: Engr. Aamir Jamil (Lead AI, Robotics, STEAM & CEH Trainer), Miss Zia Ul Ain (Supportive Trainer) & Awais Paracha (Manager)

Duration: 2-Day Intensive Masterclass (4 Hours/Day = 8 Hours Total)

Power Source: Safe Low-Voltage AA Batteries

User Manual: Detailed step-by-step guide with diagrams

Applied Expansion: Out-of-kit rovers, vibro-bots & arena trials



10+	8	2 Days
Builds & Trials	Learning Hours	Mastery Track

✓ Zero Coding ✓ No Arduino ✓ No Soldering

ACCREDITED TEAM

Meet the Instructional & Management Team

Expert educators and programme management from CS Technologies specializing in experiential STEAM, robotics, AI, and child-safe hardware prototyping.

LEAD TRAINER

Engr. Aamir Jamil

Lead AI, Robotics, STEAM & CEH Trainer

CS Technologies

Experienced, professional AI and robotics trainer and lead with comprehensive expertise across all STEAM disciplines. Dedicated to demystifying electronics, intelligent automation, and child-safe robotics for young learners through hands-on project creation and intuitive scientific experiments.

- Lead AI, Robotics & CEH
- All STEAM Disciplines
- Circuit & Sensor Prototyping
- Hands-on Engineering

SUPPORTIVE TRAINER

Miss Zia Ul Ain

Supportive Trainer

CS Technologies

Supportive trainer assisting during workshop sessions, providing hands-on student mentoring, circuitry troubleshooting, and guiding young learners through hardware assembly and safe experiment execution.

- Supportive Trainer
- Lab Assistance
- Breadboard Guidance
- Student Mentorship

MANAGER

Awais Paracha

Manager

CS Technologies

Overseeing programme operations, school coordination, student logistics, and session quality assurance. Dedicated to delivering an organized, safe, and impactful STEAM learning experience for students, parents, and academic institutions.

- Manager
- Programme Operations
- Logistics & Scheduling
- Quality Assurance

CORE COMPETENCIES

What Students Will Learn

A proven progressive pathway covering circuit theory, active electronic components, sensors, and practical troubleshooting.

Component Mastery

Identify breadboards, LEDs, resistors, switches, buzzers, tilt sensors, and DC motors by sight and function.

Circuit Principles

Understand voltage, current, resistance, LED polarity (anode/cathode), and the role of current-limiting resistors.

Sensors & Motion

Harness light sensors (LDRs) and orientation tilt switches to trigger real-world alarms and spin motor fans.

Fault Troubleshooting

Form scientific curiosity by debugging loose connections, reversed polarity, open loops, and loose jumper leads.

10 Hands-on Engineering Projects & Arena Challenges

Day 1 establishes circuit and sensor foundations through 6 core hardware labs. Day 2 expands into compound automation, biomimetic locomotion, motorized rovers, and arena time-trials!



PROJECT 1 • DAY 1 FOUNDATION

Time: 25 mins Output: Light

Mini LED Torch

Objective: Build a complete battery-powered handheld torch while mastering LED polarity and resistor protection.

COMPONENTS USED

- 2 × AA Batteries
- Battery Holder
- 1 × Red/White LED
- 1 × 330Ω Resistor
- Jumper Wires
- Cardboard Housing

BUILD PROCEDURE

1. Identify positive (+) and negative (-) battery terminals.
2. Determine LED anode (long leg) and cathode (short leg).
3. Wire battery positive to resistor, then to LED anode.
4. Connect LED cathode back to battery negative to close the circuit.
5. Mount securely into custom cardboard torch structure.

Learning: Complete circuits, LED polarity & resistor safety.



PROJECT 2 • DAY 1 FOUNDATION

Time: 25 mins Output: Controlled Light

Push-Button Light

Objective: Control an LED dynamically using an interactive push-button switch to grasp open vs. closed circuits.

COMPONENTS USED

- Mini Breadboard
- Push Button Switch
- 1 × LED
- 1 × 330Ω Resistor
- 2 × AA Battery Pack
- Jumper Wires

BUILD PROCEDURE

1. Place the push-button switch across the center divider of the breadboard.
2. Wire the switch in series with the resistor and LED.
3. Press the button to complete the loop (closed circuit) and illuminate LED.
4. Release the button to break the connection (open circuit).

Learning: Understand manual switches & current interruption.



PROJECT 3 • DAY 1 FOUNDATION

Time: 25 mins Output: Acoustic Sound (Buzzer)

Electronic Quiz Buzzer

Objective: Convert electrical energy into audible sound waves for a fun, competitive classroom game quiz panel.

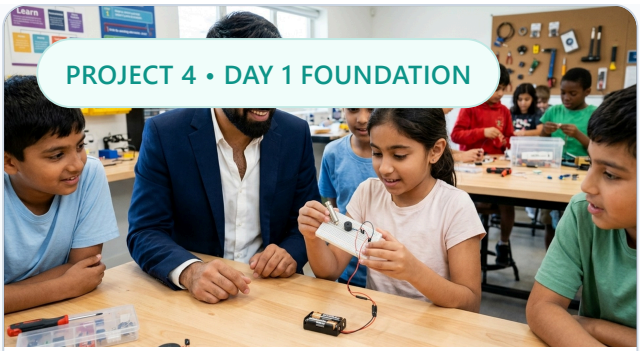
COMPONENTS USED

- Active Buzzer (Piezo)
- Tactile Push Button
- Battery Supply
- Jumper Wires
- Cardboard Quiz Console

BUILD PROCEDURE

1. Verify the active buzzer's rated polarity (+ and - leads).
2. Connect the buzzer in series with the push button and battery.
3. Press button to produce an instant audible tone.
4. Decorate and install onto a cardboard quiz contestant console.

Learning: Electrical-to-acoustic energy conversion & signals.



PROJECT 4 • DAY 1 FOUNDATION

Time: 25 mins Output: Alarm Sound

Tilt-Activated Alarm

Objective: Engineer an orientation-sensitive security alarm using a mechanical ball-tilt switch sensor.

COMPONENTS USED

- Metallic Tilt Switch
- Active Buzzer
- Mini Breadboard
- Battery Supply
- Jumper Leads

BUILD PROCEDURE

1. Mount the cylindrical tilt switch vertically on the breadboard.
2. Connect in series with the buzzer and battery power source.
3. Observe conductibility as internal metallic ball rolls with gravity.
4. Gently tilt the circuit past 45° to sound the intrusion alarm.

Learning: Sensor triggering, motion detection, and security circuits.



PROJECT 5 • DAY 1 FOUNDATION

Time: 30 mins Output: Measured/Visual Light Response

Light Detector

> **Objective:** Discover photo-sensing physics by testing how ambient light changes an LDR's resistance and affects circuit behavior.

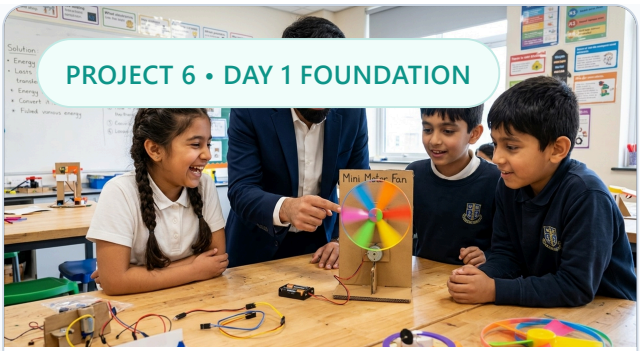
COMPONENTS USED

- Light-Dependent Resistor (LDR)
- 1 × Bright LED
- 1 × 330Ω Resistor
- 2 × AA Batteries
- Mini Breadboard

BUILD PROCEDURE

1. Observe LDR sensor surface under direct room lighting.
2. Build series circuit with LDR, protective resistor, and LED.
3. Cup hands or shade LDR to observe resistance change.
4. Observe how light variations regulate electron current.

Learning: Optical sensors, variable resistance & environmental triggers.



PROJECT 6 • DAY 1 FOUNDATION

Time: 30 mins Output: Mechanical Rotation & Airflow

Mini Motor Fan

> **Objective:** Convert electrical energy into rotational kinetic motion and aerodynamic airflow using a 3–6V DC motor.

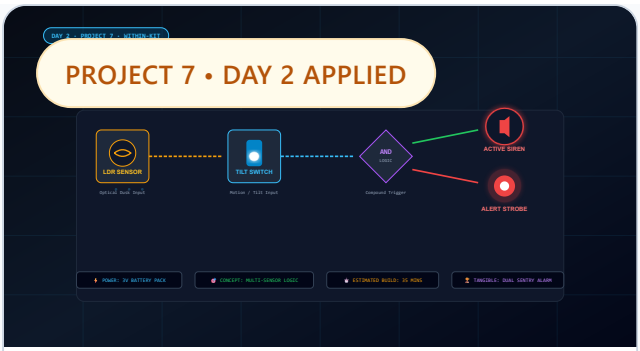
COMPONENTS USED

- Small 3–6V DC Motor
- Lightweight Propeller Blade
- Cardboard Stand
- 2 × AA Battery Pack
- Jumper Leads

BUILD PROCEDURE

1. Inspect motor terminals and verify rated supply voltage.
2. Affix DC motor securely onto a sturdy cardboard support.
3. Press-fit the lightweight plastic propeller fan blade onto shaft.
4. Connect to battery pack and feel the refreshing airflow.

Learning: Electromagnetic rotation, mechanical propulsion & aerodynamics.



Time: 35 mins Output: Multi-Sensor Alarm Siren

Dual-Sensor Smart Night Sentinel

> **Objective:** Interlink the optical LDR and gravity tilt switch on a single breadboard to engineer an autonomous anti-intruder sentry alarm.

COMPONENTS USED (WITHIN-KIT)

- 1 × LDR Sensor
- 1 × Tilt Switch
- 1 × Active Buzzer
- 1 × Red Strobe LED
- 2 × 330Ω Resistors
- Breadboard & Jumpers

BUILD PROCEDURE

1. Wire LDR in series with resistor to establish ambient dusk trigger threshold.
2. Route tilt switch in parallel to detect physical displacement or movement.
3. Bridge output rails to activate both active buzzer and red strobe LED simultaneously.
4. Shade the LDR to simulate night, then tilt the enclosure to trip the alarm siren!



Time: 35 mins Output: Autonomous Vibration Locomotion

Autonomous Vibro-Bot & Scribbler

> **Objective:** Harness an Eccentric Rotating Mass (ERM) on a DC motor to convert rotational torque into autonomous directional skittering.

COMPONENTS USED (KIT + MECHANICAL BASE)

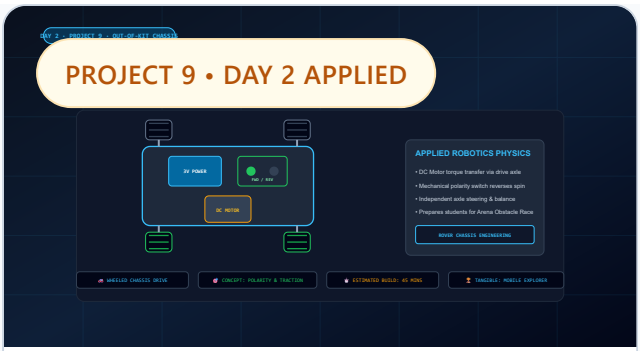
- DC Motor (3–6V)
- Eccentric Counterweight Cam
- Bristle / Skate Chassis
- 2 × AA Battery Pack
- Marker Tripod Tips

BUILD PROCEDURE

1. Affix DC motor onto the angled bristle chassis platform using foam tape.
2. Mount the eccentric off-center counterweight cam onto the high-speed motor spindle.
3. Power the circuit and observe high-frequency vibration harmonics.
4. Tune bristle tilt angles to propel the robot forward and attach markers for self-drawing art.

- Learning: Multi-sensor conditional logic, compound routing & automated

- Learning: Eccentric Rotating Mass (ERM), mechanical vibration physics &



Time: 45 mins Output: Wheeled Forward/Reverse Drive

Motorized Dual-Drive STEM Rover

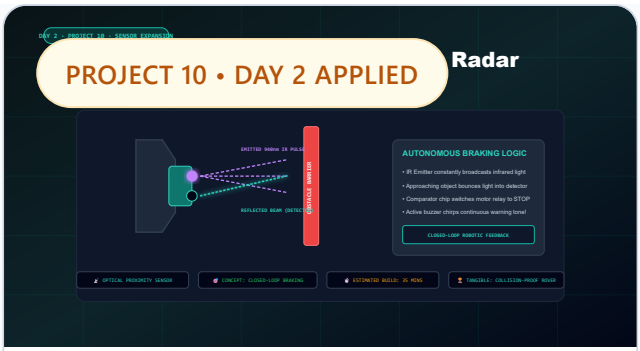
> **Objective:** Construct a wheeled robotic rover chassis with polarity-reversing switches to navigate forward, reverse, and execute turns.

COMPONENTS USED (OUT-OF-KIT HARDWARE CHASSIS)

- Chassis Baseplate
- 4 × Rubber Grip Wheels
- 2 × Steel Axles & Bushings
- DC Drive Motor & Pulley
- DPDT Polarity Switch

BUILD PROCEDURE

1. Snap axle bushings and 4 high-traction rubber wheels onto the chassis plate.
2. Mount DC drive motor and align the transmission band to the rear drive axle.
3. Wire the DPDT switch to reverse DC polarity when toggled.
4. Test rover forward and reverse navigation and calibrate straight-line tracking.



Time: 35 mins Output: Proximity Alert & Auto-Braking

IR Autonomous Obstacle Radar

> **Objective:** Integrate an active infrared proximity radar module to give rovers autonomous collision-avoidance braking and proximity chirps.

COMPONENTS USED (MODULAR EXPANSION)

- IR Proximity Module
- 940nm Emitter & Photodiode
- Active Buzzer Warning
- Motor Intercept Relay
- Breadboard Jumpers

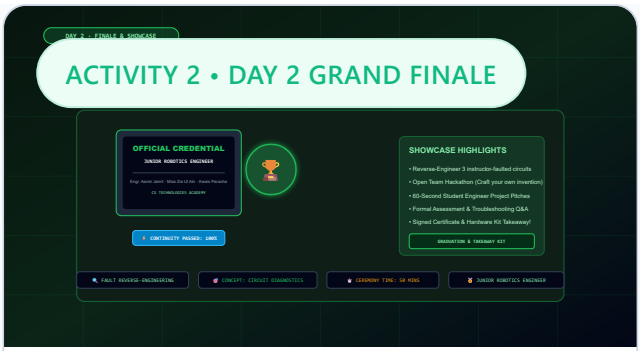
BUILD PROCEDURE

1. Mount the dual-lens IR optical sensor module at the front bumper of the rover.
2. Connect VCC (3V), Ground, and output signal pins to the breadboard logic line.
3. Calibrate onboard sensitivity potentiometer to achieve 10–15 cm detection distance.
4. Drive rover toward an obstacle to observe automatic motor cutoff and audible proximity chirps!

- Learning: Motor polarity reversal (H-bridge concept), mechanical drive trains & rover dynamics.

- Learning: Optical infrared reflection, closed-loop feedback control & collision avoidance.





Time: 45 mins Format: Team Obstacle Time-Trial

Grand "Mars Rover" Obstacle Arena

> **Objective:** Student engineering teams deploy calibrated rovers onto a 3-meter arena track featuring slalom gates, 15° incline ramps, and speed trials.

ARENA HARDWARE EQUIPMENT

- 3-Meter Arena Track Mat
- Slalom Gate Pylons
- 15° Terrain Incline Ramp
- Stopwatch Timer
- Team Relay Badges

CHALLENGE PROTOCOL

1. Inspect track layout, measure incline angles, and calibrate rover tire traction.
2. Complete 2 practice laps to adjust battery ballast and driver switch reactions.
3. Official Timed Trial: navigate slalom gates without knocking course pylons.
4. Summit the incline ramp and cross the finish line to register team lap time!

- Learning: Applied team engineering, live prototyping iteration & mechanical stress testing.

Time: 50 mins Format: Reverse-Engineering & Showcase

Speed-Wire Diagnostics & Showcase

CURRICULUM ROADMAP

Structured 2-Day Masterclass Roadmap (8 Hours Total)

Balanced across 2 progressive 4-hour days combining foundational electronic circuit theory, multi-sensor compound automation, out-of-kit motorized rovers, and grand arena speed trials.

DAY 1 FOUNDATION

Duration: 4 Hours (Morning + Afternoon)

Core Circuit Engineering & Sensor Hardware

Theory & Concepts (6 Foundation Builds):

- Atomic electricity, voltage, current loops & breadboard bus architecture
- LED diode polarity & current-limiting resistor protection (Ohm's Law intuition)
- Momentary tactile switches & series interruption logic
- Acoustic transducers & piezoelectric sound wave generation
- Gravitational tilt sensing & mercury-free mechanical

DAY 2 APPLIED ROBOTICS

Duration: 4 Hours (Morning + Afternoon)

Autonomous Robotics, Multi-Sensors & Arena Trials

Theory & Concepts (Applied Vehicles & Challenges):

- Compound sensor logic: Series/parallel multi-sensor integration
- Eccentric Rotating Mass (ERM) kinetic oscillation & bristle locomotion
- Wheeled robot chassis mechanics, axle alignment & high-traction tires
- Directional polarity reversal using DPDT rocker switching

rolling ball contacts
- Light-dependent cadmium sulfide photoresistors (LDR) & optical variance
- Direct-current (DC) electromagnetic motors & propeller airflow aerodynamics
- Diagnostic fault isolation & "Silent Circuit" reverse-engineering drill

- Infrared (IR) 940nm optical reflection, proximity sensing & auto-braking
- Tactical track navigation, slalom steering & ramp climb physics
- Competitive reverse-engineering diagnostics under timed pressure
- Junior Robotics Engineer portfolio showcase & certification

Timeline	Workshop Activity & Build Target
`00–20 min`	Welcome, Lab Safety Orientation & Solderless Tool Handling
`20–55 min`	**Project 1:** Mini LED Torch (Solderless Loop & Polarity)
`55–90 min`	**Project 2:** Push-Button Light (Loop Interruption & Control)
`90–125 min`	**Project 3:** Electronic Quiz Buzzer (Acoustic Signaling)
`125–140 min`	Hydration Break & Component Sort Inspection
`140–170 min`	**Project 4:** Tilt-Activated Motion Alarm (Gravitational Trigger)
`170–200 min`	**Project 5:** Light Detector (LDR Optical Variance Testing)
`200–225 min`	**Project 6:** Mini Motor Fan

Timeline	Workshop Activity & Engineering Challenge
`00–20 min`	Day 1 Review, Robotics Systems Overview & Chassis Physics Briefing
`20–55 min`	**Project 7:** Dual-Sensor Smart Night Sentinel (LDR + Tilt compound logic)
`55–95 min`	**Project 8:** Autonomous Vibro-Bot (ERM vibration locomotion & bristle skates)
`95–135 min`	**Project 9:** Motorized Dual-Drive STEM Rover (Wheeled chassis & polarity drive)
`135–150 min`	Pit Stop: Drivetrain Calibration & Battery Power Optimization
`150–185 min`	**Project 10:** Infrared (IR) Autonomous Obstacle Radar Module

Timeline	Workshop Activity & Build Target
	(Electromechanical Propulsion)
225–240 min	Day 1 Diagnostic Challenge & "Silent Circuit" Reverse-Engineering Drill

Timeline	Workshop Activity & Engineering Challenge
185–225 min	**Grand Arena:** The 3-Meter "Mars Rover" Obstacle Course & Slalom Grand Prix
225–240 min	Speed-Wire Reverse-Engineering Relay, Peer Showcase & Certification Ceremony

HARDWARE BILL OF MATERIALS

Master Hardware & Robotics Expansion Pack

Every student receives a complete, re-usable hardware kit with safe, solder-free components plus out-of-kit chassis assemblies for Day 2 robotics challenges.

Filter hardware components...

20+ Complete Hardware Items & Arena Materials

Component Specification	Educational & Robotics Purpose	Quantity
📦 **PART A: DAY 1 FOUNDATION HARDWARE & SENSOR KIT**		
Mini Breadboard	Solderless circuit prototyping platform	1
LEDs (Mixed Colours)	Visual output indicators, polarity training	6
Resistors (220Ω, 330Ω, 1kΩ)	Current-limiting and circuit protection	10
2 × AA Battery Holder	Safe, low-voltage power supply enclosure with switch	1
AA Batteries (Alkaline)	Portable DC power source (3V safe nominal)	2
Jumper Wires (Flexible Male-Male)	Rapid solderless circuit connections	10
Push Buttons (Tactile)	Manual switching and signal triggering	2
Active Buzzer	Audible alert and quiz sound signaling	1
Crocodile Clips	Firm mechanical and electrical terminal gripping	2
Small DC Motor (3–6V)	Electromechanical kinetic rotation & propeller drive	1

Component Specification	Educational & Robotics Purpose	Quantity
Light-Dependent Resistor (LDR)	Ambient optical environmental sensing	`1`
Tilt Switch Sensor	Motion and gravitational tilt detection	`1`
Component Storage Box & Cadet Labels	Individual kit organization and student accountability	`1`
Illustrated Student Guidebook & Manual	Step-by-step schematics, pictorial troubleshooting & DIY blueprints	`1`
🤖 **PART B: DAY 2 APPLIED ROBOTICS, CHASSIS & ARENA EXPANSION PACK**		
Modular Lightweight Robot Chassis & Motor Mount	Pre-drilled mounting plate for dual-drive rover assembly	`1 Set`
High-Traction Rubber Drive Wheels & Front Castor	Low-friction ground contact and stable 3-point rover balance	`2 Wheels + 1 Castor`
DPDT Polarity Reversing Rocker Switch	Bi-directional forward/reverse drivetrain steering control	`1`
Eccentric Rotating Mass (ERM) Cam & Bristle Skates	Converts rotary motor spin into directional vibrational hopping (Vibro-Bot)	`1 Pack`
Infrared (IR) Proximity & Obstacle Radar Sensor Module	940nm optical reflection detector with onboard comparator & sensitivity pot	`1 Module`

Component Specification	Educational & Robotics Purpose	Quantity
Competition Arena Pack (3m Track Mat, Slalom Cones, 15° Ramp)	Tactical trial course for timed navigation, obstacle avoidance & hill climbs	`Arena Facility`

> **Included User Manual & Multi-Project Potential:** Every student receives a comprehensive, professionally illustrated manual. Beyond the 10 masterclass builds, students can follow the manual to build many additional DIY experiments at home (such as Morse code transmitters, multi-LED logic circuits, vibration sensors, and light-sensitive alarms) using the same kit hardware!

> **Day 2 Out-of-Kit Materials Included:** The rover chassis plates, rubber traction wheels, ERM vibration units, IR optical radar modules, and 3-meter competition arena track are fully supplied by CS Technologies for all Day 2 activities.

⚠ Safety Guidelines

- Use strictly low-voltage safe battery power (3V nominal).
- Always disconnect power before re-routing breadboard connections.
- Always use an appropriate current-limiting resistor with each LED.
- Secure DC motors and fan propellers firmly before applying current.
- Keep hair and loose items clear from spinning propeller blades.
- Continuous adult mentoring provided throughout all activities.

Expected Outcomes

- Draw and explain basic schematic circuit loops and polarities.
- Explain why current-limiting resistors prevent LED burnout.
- Demonstrate manual switching on both series and open configurations.
- Calibrate environmental response using tilt and light sensor circuits.
- Independently detect and resolve at least one real circuit fault.

Track / Day	Project or Engineering Challenge	Primary STEM Concept	Tangible Output
DAY 1	**1. Mini LED Torch**	Complete Circuit Loop & LED Polarity	Controlled Directional Illumination
DAY 1	**2. Push-Button Light**	Mechanical Switching & Interruption	Interactive Light Toggle Control
DAY 1	**3. Electronic Quiz Buzzer**	Electrical-to-Sound Energy Conversion	Audible Game Tone Signaling
DAY 1	**4. Tilt-Activated Alarm**	Gravitational Orientation Sensing	Motion-Activated Security Siren
DAY 1	**5. Light Detector**	Light-Sensitive Resistance (LDR)	Optical Sensor Modulation & Feedback
DAY 1	**6. Mini Motor Fan**	Electrical-to-Mechanical Energy Conversion	High-Speed Propeller Airflow & Thrust
DAY 2	**7. Smart Night Sentinel**	Compound Dual-Sensor Logic (LDR + Tilt)	Automated Anti-Theft Night Alert
DAY 2	**8. Autonomous Vibro-Bot**	ERM Kinetic Oscillation & Vibration Dynamics	Biomimetic Bristle-Skate Hopping Locomotion
DAY 2	**9. Motorized STEM Rover**	Drivetrain Mechanics & Polarity Reversal	Direction-Controlled Wheeled Vehicle
DAY 2	**10. IR Obstacle Radar**	940nm Proximity Reflection & Auto-Braking	Autonomous Proximity Collision Avoidance

Track / Day	Project or Engineering Challenge	Primary STEM Concept	Tangible Output
DAY 2 ARENA	**Activity 1: Mars Rover Arena**	Obstacle Slalom, 15° Incline Ramp & Precision Racing	3-Meter Timed Tactical Arena Trials
DAY 2 ARENA	**Activity 2: Speed-Wire Relay**	Circuit Diagnostics, Rapid Assembly & Team Hackathon	Junior Robotics Engineer Certification